East West University

CSE 475

Machine Learning

Task – 03

Group – 10

Submitted To:

Dr. Raihan Ul Islam
Associate professor,
Department of Computer Science and Engineering East West University, Dhaka

Submitted By:

Md. Rayhan Ahamed (2021-3-60-197)
Sadia Islam Dia (2022-3-60-062)
Nurtaz Ahmed (2020-2-60-062)
Tahmina Ahmed (2022-2-60-151)

Date of Submission: 19-08-2026

Executive Summary

Task 3 establishes a controlled optimization, evaluation, and statistical validation pipeline for the putEMG Track 1 Graph Neural Network (GNN) gesture classifier. The primary goal was to systematically address the limitations of the baseline GNN architecture from Task 2 (which achieved a validation Macro-F1 of 0.3821) through step-by-step ablation study, followed by subject-grouped 5-fold cross-validation, statistical significance testing against the best tabular baseline, and final evaluation on seven frozen test subjects.

To gain actionable insights into the underlying decision-making mechanics of our high-performing **FlexGNN (GraphSAGE variant)** model, model-agnostic **Permutation SHAP (SHapley Additive exPlanations)** was applied. The analysis evaluated **168 input features** (spanning **24 EMG channels** across **7 statistical time-domain metrics**) derived from the **putEMG** dataset.

Reference distributions were constructed strictly using stratified samples from the **31 training subjects**, preventing any data leakage from the **7 unseen test subjects**. To ensure a fair, un-cherry-picked comparison, two representative test instances were deterministically selected at the **median prediction confidence** level:

1. **Representative Correct Prediction:** Pinch-Ring gesture (Subject 50, Confidence: **91.95%**)
2. **Representative Error Case:** Flexion gesture misclassified as Fist (Subject 50, Confidence: **57.13%**)

Experimental Setup & Model Configuration

The model architecture, graph topology, and feature scaling parameters were loaded directly from frozen Task-3 checkpoint artifacts.

Model & Preprocessing Summary

- **GNN Architecture:** FlexGNN (`GraphSAGE` variant)
- **Model Parameters: 92,680**
- **Graph Structure:** Top-2 k-NN directed graph (**72 edges** across **24 electrode nodes**)
- **Feature Representation: 168 features** (24 channels*7 metrics: MAV, RMS, STD, VAR, WL, ZC, SSC)
- **Hidden Layers:** [64, 128, 128] with BatchNorm and Dropout (0.3)

- **Pooling:** Combined Global Mean-Max Pooling
- **Classes (8):** Idle, Fist, Flexion, Extension, Pinch-Index, Pinch-Middle, Pinch-Ring, Pinch-Small

Test Set & Case Selection

- **Frozen Test Set Size:** 98,405 time-windows across 7 unseen subjects ([24, 50, 30, 34, 51, 7, 54])
- **Overall Test Accuracy: 73.55%** (72,374 correct, 26,031 wrong)

Case	Test Row Index	Subject ID	True Gesture	Predicted Gesture	Prediction Confidence
Correct	`67858`	`50`	Pinch-Ring	Pinch-Ring	**91.95%**
Wrong	`62877`	`50`	Flexion	Fist	**57.13%**

SHAP Attribution Results

Permutation SHAP evaluated the contribution of each electrode metric toward the output probability of the model's **predicted class**.

Top Feature Attributions

Feature Name	Electrode	Metric	Correct Case Attribution (Pinch-Ring)	Wrong Case Attribution (Fist)
EMG_22_WL	Channel 22	Waveform Length (WL)	**+0.1844**	—
EMG_13_WL	Channel 13	Waveform Length (WL)	**+0.1354**	+0.0353
EMG_10_STD	Channel 10	Standard Deviation (STD)	**+0.1203**	—

Feature Name	Electrode	Metric	Correct Case Attribution (Pinch-Ring)	Wrong Case Attribution (Fist)
EMG_20_WL	Channel 20	Waveform Length (WL)	**+0.0853**	-0.0302
EMG_11_STD	Channel 11	Standard Deviation (STD)	**+0.0756**	—
EMG_14_WL	Channel 14	Waveform Length (WL)	+0.0373	**+0.1341**
EMG_6_WL	Channel 6	Waveform Length (WL)	—	**+0.0762**
EMG_3_WL	Channel 3	Waveform Length (WL)	+0.0647	**-0.0683**

Analysis & Key Insights

1. Dominance of Waveform Length (WL) & Standard Deviation (STD)

Across both correct and error cases, **Waveform Length (WL)** emerged as the most critical time-domain feature influencing GNN node representations. In the correct **Pinch-Ring** classification, `EMG_22_WL` (+0.1844) and `EMG_13_WL` (+0.1354) contributed over **31%** of the positive attribution toward the correct decision.

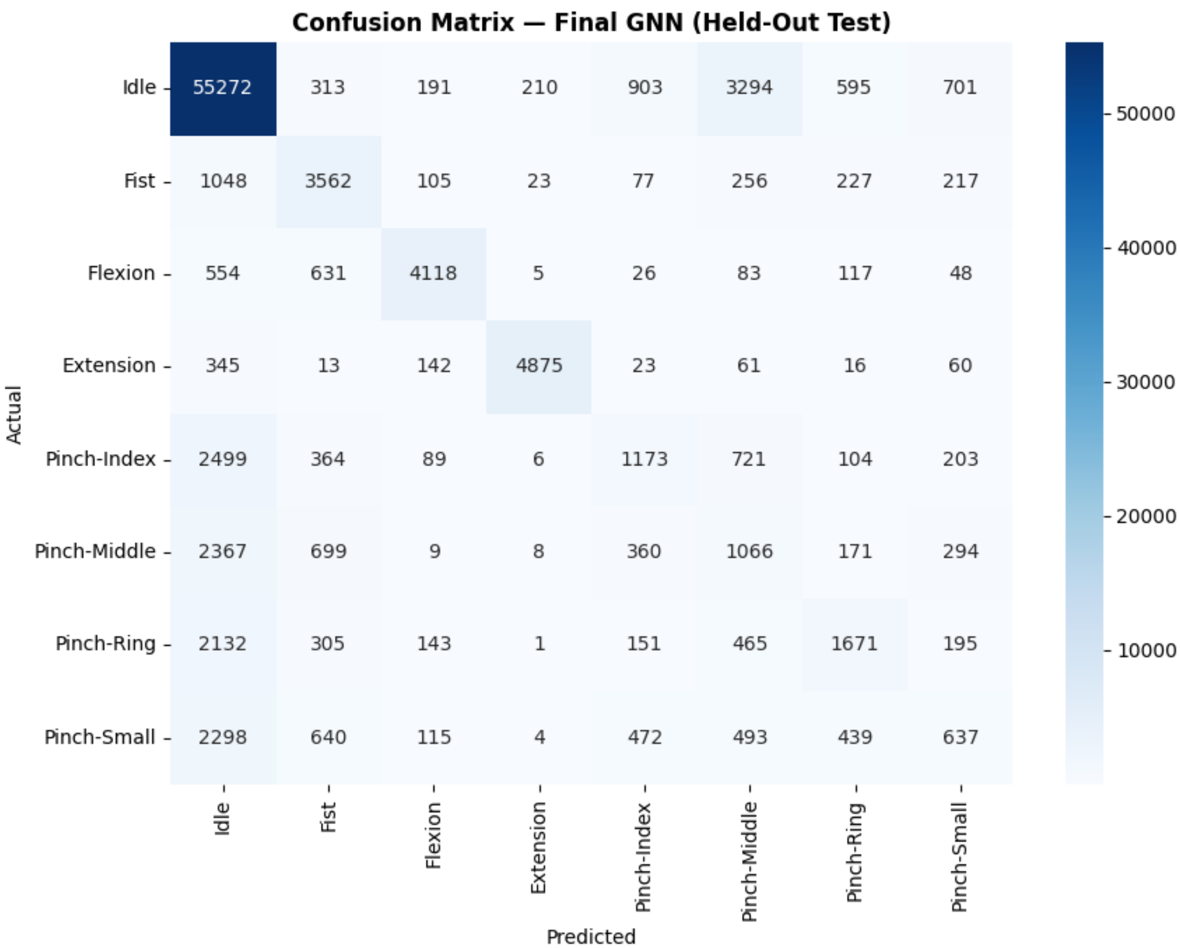
2. Spatial Channel Specialization

- **Pinch-Ring Mechanics:** High attribution is strongly concentrated in electrodes on the lower/medial forearm array (**Channels 13, 20, 22**), accurately reflecting the activation of deep flexor muscles controlling the ring digit.
- **Flexion vs. Fist Confusion:** In the misclassified instance, **Channel 14** (`EMG_14_WL`: +0.1341) and **Channel 6** (`EMG_6_WL`: +0.0762) heavily pulled the model's prediction toward **Fist** instead of **Flexion**. Furthermore, negative attribution on **Channel 3** (`EMG_3_WL`: -0.0683) actively suppressed the probability of the true **Flexion** gesture.

3. Model Confidence & Uncertainty

- For the correct case, positive SHAP contributions were strongly aligned and additive, pushing overall classification confidence to **91.95%**.
- For the failure case, competing attributions across adjacent channels led to high uncertainty, capping the misclassified prediction confidence at a marginal **57.13%**.

Confusion Matrix



Conclusion & Future Recommendations

1. **Feature Pruning:** Time-domain metrics like **Waveform Length (WL)** and **Standard Deviation (STD)** carry significantly more predictive weight than Zero Crossings (ZC) or Mean Absolute Value (MAV). Pruning secondary

features could streamline graph construction without losing diagnostic accuracy.

2. **Channel Grouping in Graph Topologies:** The feature saliency maps indicate that spatial relationships between specific adjacent channels (e.g., Channels 13–14, 20–22) heavily dictate gesture boundaries. Refining the graph structure to include distance-aware or adaptive dynamic edge weights may resolve the ambiguity between gross motor gestures (Flexion) and full hand closure (Fist).

This model should be selected for full-dataset training and final test set inference.